Cash App Smart Warnings

Rethinking scam warnings to make payments safer without slowing them down



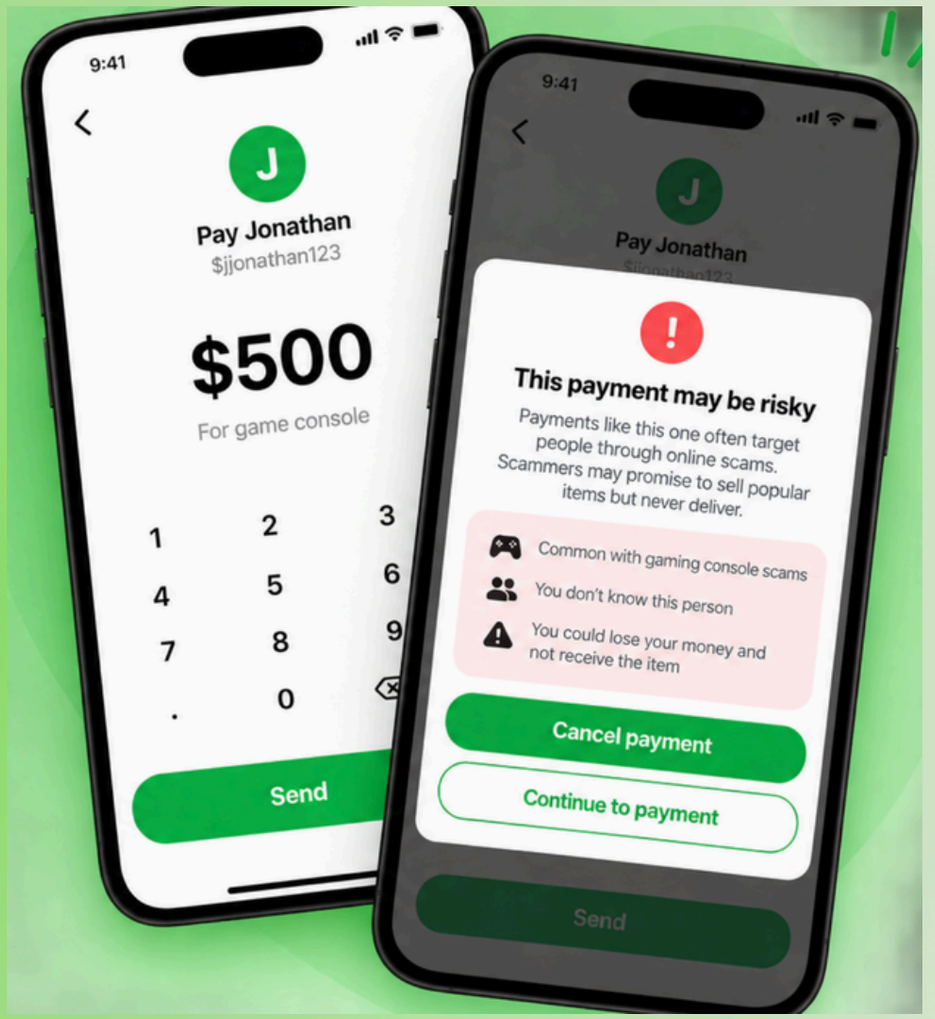
How can we make digital payments safer without making them slower?

- Peer-to-peer (P2P) payment apps like Cash App have made sending money almost effortless. Whether splitting bills, paying rent, or purchasing something online, users can transfer money in seconds.
- But that same convenience creates opportunities for scammers.
- Cash App already uses machine learning to identify suspicious transactions and warn users before payments are completed.
- According to Cash App, customers cancel or abandon warned payments nearly 50% of the time. However, many warned payments still proceed, and not all of them are fraudulent.

The challenge isn't just detecting a risky payment. It's helping users make the right decision after receiving a warning.

About This Case Study

- In this case study, I'll explore Cash App's existing scam prevention experience, examine why users might proceed despite warnings, and propose a solution that makes those warnings more actionable without introducing unnecessary friction.
- I'll then define an MVP, success metrics, and an experimentation strategy to evaluate the proposed solution.



Hypothesis

If Cash App introduces contextual, risk-based payment warnings, then confirmed scam payments completed after a warning will decrease without reducing legitimate payment completion by more than 1 percentage point.
Source: Cash App — How AI-Driven Scam Warnings Help Prevent Fraud

What Can We Learn From Competitors?

How do payment apps protect users from scams?

Cash App, Venmo, and Zelle all face a similar challenge: helping users recognize suspicious payments before money is transferred. However, their approaches to scam prevention differ.

	Cash App	Venmo	Zelle	Smart Warnings (proposed)
Risk Detection	ML-powered detection	Fraud detection systems	Bank-dependent	Uses Cash App's existing detection
Warnings	In-app scam alerts	Risk-based alerts on Friends & Family payments	Varies by bank	Context-specific, tiered by risk
Explains why it's risky?	Not clear from public materials	Describes risk level	Varies by bank	Yes: names the likely scam pattern
Protection	Certain scams may qualify for reimbursement	Limited protection for personal payments	Limited recourse for authorized scams	Adds a cooling-off hold for high-risk payments

What Are the Common Themes?

1. Detection comes first.

Payment platforms use fraud detection systems to identify suspicious transactions.

2. Warnings create a decision point.

Users may need to reconsider a payment before completing it.

3. Context matters.

A warning is more useful when users understand the potential risk and what they can do about it.

4. More friction isn't always better.

Additional security steps can protect customers but may also interrupt legitimate payments.

Research Takeaway

The opportunity isn't necessarily detecting more scams. It's helping users make better decisions when a payment is flagged. For Cash App, this means exploring contextual explanations, actionable verification guidance, and interventions proportionate to risk.

Who Uses Cash App?

Who are the users we're designing for?
Cash App serves people who want a fast and convenient way to send, receive, and manage money.
For this case study, we'll focus on U.S. consumers using peer-to-peer (P2P) payments.

Main Use Cases

- Sending money to friends and family
- Splitting bills and shared expenses
- Paying rent or roommates
- Purchasing items from individual sellers
- Paying for services or deposits

Risky Payment Moments

- Unfamiliar Recipients: Difficult to verify someone's identity.
- Time Pressure: Urgency can cause users to overlook warning signs.
- Warning Fatigue: Repeated alerts may lead users to ignore warnings.

Our Target Users

U.S. Cash App users aged 18–30 making payments to unfamiliar recipients.
This is a proposed research segment, not a claim about Cash App's highest-risk demographic.
We'll initially focus on users making payments for rentals, online purchases, and services.

Why This Segment?

For users: These transactions may involve unfamiliar recipients, limited verification, and pressure to pay quickly.
For Cash App: Scam losses can create reimbursement costs, additional customer support needs, and reduced customer trust.

Our Research Focus

We want to understand:

- What makes users trust unfamiliar recipients?
- Why might someone proceed despite a scam warning?
- What information would help them reconsider?
- When does additional verification become frustrating?

Key Insight

An unfamiliar recipient isn't necessarily a scammer.
Our solution must help protect at-risk users without unnecessarily interrupting legitimate payments.

What Are the Pain Points These Users Need Addressed?

At-Risk Senders

- Unclear Warnings: Users may not understand why a payment is flagged.
- Urgency: Scammers pressure users to send money quickly.
- Established Trust: Users may trust the recipient more than the app's warning.
- Lack of Verification: Users may not know how to check whether a payment request is legitimate.

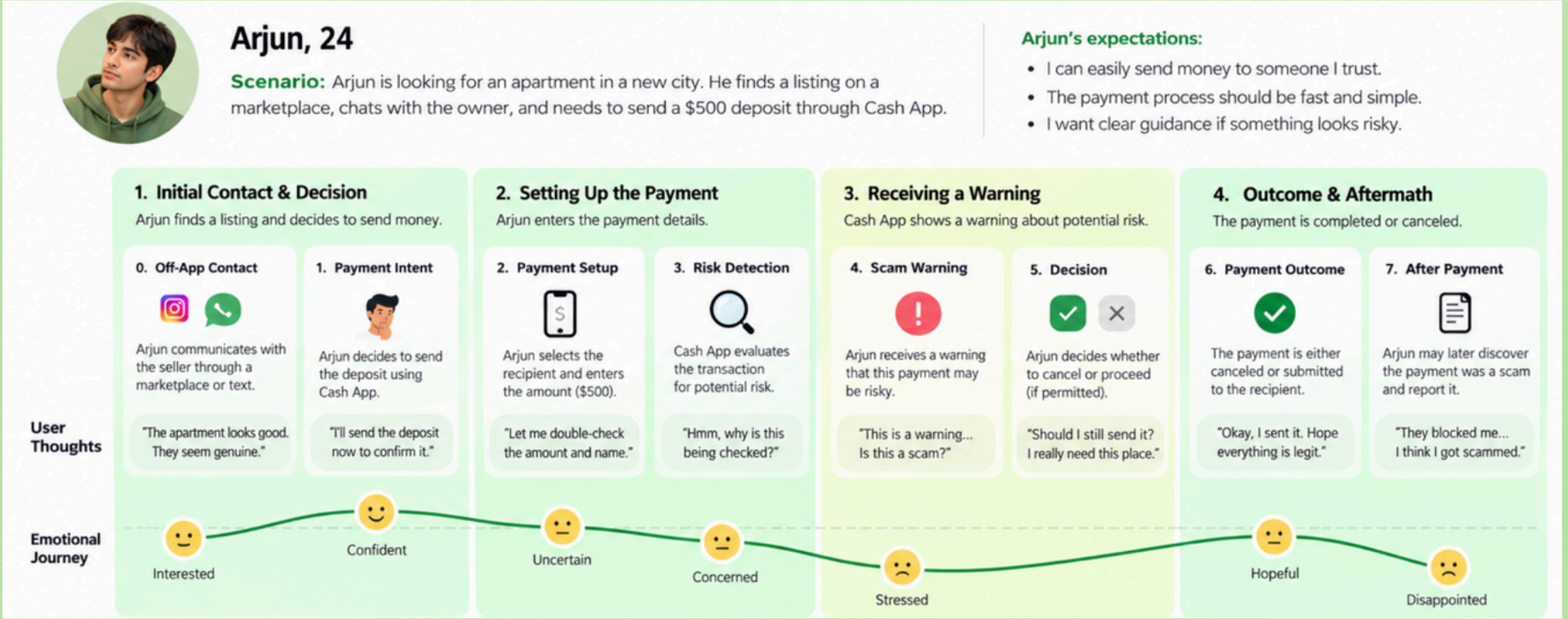
Legitimate Senders

- False Alarms: Unnecessary warnings can reduce trust in security alerts.
- Payment Friction: Extra verification steps can delay legitimate payments.
- Warning Fatigue: Repeated warnings may cause users to dismiss future alerts.

The Core Problem

How can Cash App help users recognize genuine scam risks without creating unnecessary friction for legitimate payments?

User Journey : What Happens When Cash App Flags a Payment? How does a user experience a potentially risky payment?



Where Does the Experience Break Down?

- Missing Context: Cash App may not know what happened before the payment.
- Unclear Risk: Users may not understand why their payment is suspicious.
- Trust Conflict: Users may trust the recipient more than the warning.
- Limited Guidance: Users may not know how to verify the payment request.

Research Takeaways

Based on our conceptual user journey, we identified three key opportunities:

- Missing Context: Cash App may not know the purpose of a payment or the circumstances behind it.
- Unclear Warnings: Users may not understand why a transaction was flagged as suspicious.
- Decision Uncertainty: Users may lack actionable guidance to decide whether to cancel or proceed.

The Problem

Users may receive scam warnings without enough context to confidently distinguish risky payments from legitimate ones.

This creates two potential problems:

- For at-risk users: They may ignore warnings and complete scam payments.
- For legitimate users: Unnecessary friction may interrupt everyday transactions.

The Goal

Help users make safer payment decisions without sacrificing the speed and convenience of Cash App.

We aim to:

- Provide clear, contextual scam warnings.
- Offer actionable verification guidance.
- Introduce stronger interventions for higher-risk payments.
- Minimize friction for legitimate transactions.

How Might We?

How might we help Cash App users recognize and avoid potential scams while keeping legitimate payments fast and effortless?

Success Criteria

- Reduce confirmed scam losses from flagged payment attempts by at least 10%.
- Maintain legitimate payment completion within 1 percentage point of the existing experience.

What Should Be Included in the MVP?

Building a comprehensive scam prevention system would require significant technical resources. Since Cash App already uses machine learning to identify suspicious transactions, our MVP will focus on improving how users respond to scam warnings, rather than building a new fraud detection system.

The MVP will include four core features because:

- They allow us to test whether contextual warnings help users make safer payment decisions.
- They introduce stronger protection for high-risk payments without disrupting everyday transactions.
- They help us measure whether scam losses decrease without significantly affecting legitimate payment completion.

To keep the scope manageable, features such as AI analysis of off-app messages and listings, advanced recipient verification, and a new fraud detection model will be deprioritized.

MVP User Stories

- **As a Cash App user sending money to someone new, I want warnings that fit my situation, so that I can judge whether this payment is safe.**

When a payment is flagged as medium or high risk, users select what it's for: rent or deposit, item from a seller, job or service, friend or family, or other. This tailors the warning but never lowers the system-assigned risk level.

- **As a Cash App user, I want to understand why a payment might be risky, so that I can decide based on more than a generic alert.**

Users see a contextual warning that explains the potential scam pattern, gives one specific verification tip, and offers Cancel payment as the primary action.

- **As a Cash App user making a high-risk payment, I want a chance to pause, so that pressure from the recipient doesn't rush my decision.**

Eligible high-risk payments enter a proposed 10-minute cooling-off period. Users can cancel during the hold but cannot skip it.

- **As a Cash App user whose payment is on hold, I want to see its status, so that I stay in control of my money.**

A confirmation screen shows the amount, recipient, scheduled send time, and status, with the option to cancel anytime before it's sent.

MVP Objective

Validate whether contextual, risk-tiered warnings reduce confirmed scam losses by at least 10% versus control, without reducing legitimate payment completion by more than 1 percentage point.

Final Solution : Cash App Smart Warnings

- **Payment Context**

Users select what they're paying for to receive relevant warnings.

- **Smart Warning**

Contextual scam alerts explain potential risks and provide verification tips.

- **Payment Hold**

High-risk payments enter a 10-minute cooling-off period, allowing users to reconsider or cancel.

- **Confirmation**

Users see the payment amount, recipient, status, and scheduled send time.

Key Design Principle

Make risky payments safer without slowing down legitimate ones.

What risks would Cash App face by introducing Smart Warnings?

Asking Users for Payment Context

- Adding an extra step before sending money could create friction and frustrate legitimate users.
- Users may select incorrect payment categories, resulting in irrelevant warnings.
- Scammers could instruct victims to select misleading categories to avoid suspicion.

To mitigate this, user-selected categories will never lower the system-assigned risk level.

Introducing Contextual Scam Warnings

- Providing specific warnings could reveal information about Cash App's fraud detection approach.
- Repeated warnings could lead to warning fatigue, causing users to ignore important alerts.
- Inaccurate explanations could reduce user trust in Cash App's security systems.

We must ensure warnings are clear, accurate, and relevant without exposing sensitive detection logic.

Adding a 10-Minute Payment Hold

- Delaying legitimate payments could negatively affect the user experience.
- Users may become frustrated when they cannot immediately send money.
- Scammers may redirect victims to alternative payment platforms.

To minimize disruption, payment holds will apply only to eligible high-risk transactions.

Technical Complexity & Resources

- Smart Warnings would require integration with Cash App's existing fraud detection system.
- Payment holds introduce additional complexity around scheduling, cancellation, and transaction processing.
- Building and maintaining these features would require engineering, security, and compliance resources.

Investing in Smart Warnings also means fewer resources available for other product improvements.

Measuring Success

Our goal isn't to maximize canceled payments. Success means preventing scam losses while allowing legitimate payments to continue without unnecessary friction.

Primary Metric

Confirmed Scam Losses per $1M in Flagged Payment Attempts

All flagged payment attempts count, whether completed, canceled, or held (intent-to-treat).

Target: Reduce confirmed scam losses by at least 10% compared with the existing experience.

Secondary Metrics

- Scam Completion Rate: Are fewer confirmed scams getting through?
- Warning Cancellation Rate: Are users reconsidering risky payments?

- Context Question Completion: Are users engaging with the experience?
- Payment Hold Cancellation: Does the 10-minute delay change decisions?
- Repeat Warning Overrides: Are users becoming desensitized?

More cancellations do not necessarily mean fewer scams.

Guardrail Metrics

- Legitimate Payment Completion: Must not decrease by more than 1 percentage point.
- False Positives: Minimize unnecessary interruptions.
- Payment Speed: Avoid excessive delays.
- Support Complaints: No material increase in complaints.
- Payment Reliability: No material increase in failures.

Experiment Strategy

- Control: Existing Cash App warning experience.
- Variant: Smart Warnings with contextual guidance and risk-based holds.
- Testing: Small safety pilot followed by a randomized A/B test.

Decision Rule

- Expand: Success targets and guardrails are met.
- Redesign or Stop: Safety or reliability guardrails are breached.
- Investigate: Results are mixed or inconclusive.

Measurement Challenge

All flagged payment attempts count toward the primary metric’s denominator, including completed, canceled, and held payments, to prevent higher cancellation rates from artificially improving results.

Go-to-Market Strategy (GTM)

How Would We Launch Cash App Smart Warnings?

Introducing new scam prevention features requires careful testing to ensure they improve security without disrupting legitimate payments.

Internal Testing

- Test contextual warnings and payment holds.
- Validate payment processing, cancellation, and reliability.
- Review security, privacy, and compliance requirements.

Limited Pilot

- Launch to a small group of eligible users.
- Monitor false positives, payment delays, and complaints.
- Gather feedback on warning clarity and usefulness.

A/B Testing

- Control: Existing Cash App warnings.
- Variant: Smart Warnings.
- Compare scam losses, legitimate payment completion, and user experience.

Gradual Rollout

- Expand only if success criteria are met.
- Monitor performance and customer feedback.
- Refine warnings based on observed results.

Key Takeaway

Launch cautiously, validate impact, and scale only when Smart Warnings demonstrate measurable improvements in payment safety.

Future Iterations

What's Beyond the MVP?

If Smart Warnings successfully reduces scam losses without disrupting legitimate payments, we can explore additional features.

AI-Powered Scam Detection Assistance

- Allow users to voluntarily check suspicious messages or marketplace listings before paying.
- Identify potential scam patterns and provide safety recommendations.

Recipient Trust Signals

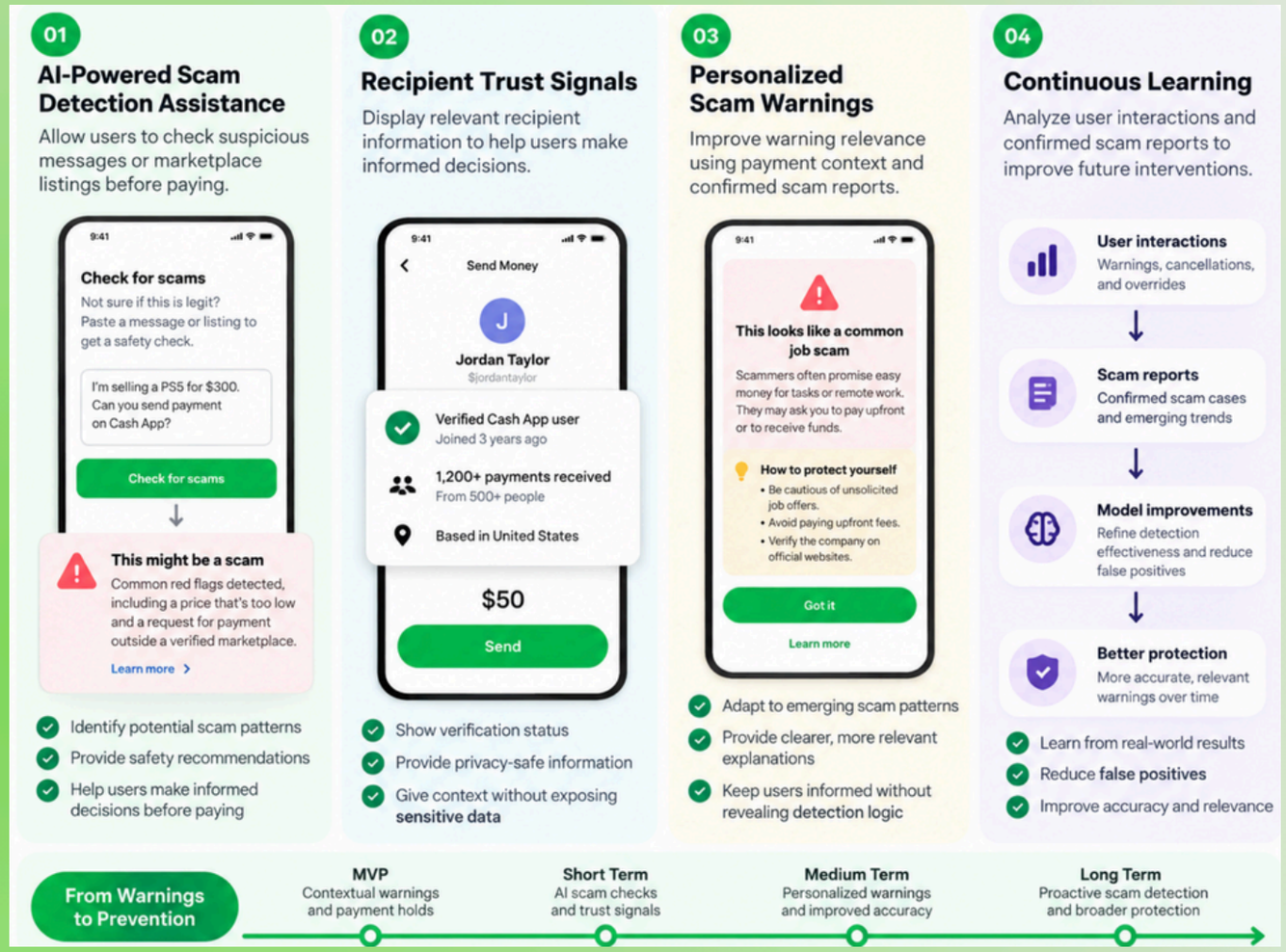
- Display relevant recipient information to help users make informed decisions.
- Explore privacy-safe indicators such as verification status and payment history.

Personalized Scam Warnings

- Improve warning relevance using payment context and confirmed scam reports.
- Adapt explanations to emerging scam patterns without revealing sensitive detection methods.

Continuous Learning

- Analyze warning interactions and subsequent scam reports.
- Refine interventions to reduce false positives and unnecessary friction.



Final Thoughts

- Cash App has already invested in detecting suspicious payments. My proposed solution, Smart Warnings, explores how the experience after detection could help users make safer decisions.
- By combining contextual explanations, risk-based interventions, and actionable guidance, the goal is to prevent scam losses without making everyday payments unnecessarily complicated.
- My background in security and threat modeling shaped this case study. I approached each product decision by thinking like an attacker: Could a scammer coach someone through the payment-context question? Could a payment delay push victims toward another platform?
- In trust and safety, a product must work for its intended users while withstanding people actively trying to exploit it.

This case proposes a solution and a way to test it. Whether it actually reduces scam losses is what the experiment would show.

My biggest learning is that building a safer product isn't about adding more warnings. It's about knowing when to intervene, how strongly to intervene, and when to stay out of the user's way.

And sometimes, the most valuable product feature is the one that convinces someone not to press Send.

Thank you for checking out this case study!